Acute meningitis

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Introduction

- CNS infections are important causes of morbidity and mortality world wide
  - Infection occurs in anatomically confined space
  - The CNS is an immunodeficient site
  - Penetration of drugs in to CSF is hampered by the Blood Brain Barrier

- **Acute meningitis**
  - Acute bacterial meningitis
  - “Aseptic” viral meningitis
- **Chronic meningitis**
  - Neurological symptoms/ CSF abnormalities persist for > 4 weeks

Acute Bacterial Meningitis

Cont…

- **Definition**
  - Bacterial meningitis is an acute purulent infection within the subarachnoid space
  - The meninges, the subarachnoid space, and the brain parenchyma are all frequently involved in the inflammatory reaction (meningoencephalitis)

Pathophysiology and pathogenesis

- **Meningeal invasion**
  - Pathogens possess surface components, such as fimbriae or pili, that enhance mucosal colonization
  - Colonization of the host mucosal epithelium is facilitated by evasion of mucosal secretory IgA through pathogen secretion of IgA protease
  - IgA protease inactivates the mucosal antibody and facilitates bacterial attachment to host epithelial cells.
  - Traverse Blood Brain Barrier, BBB,
  - Choroid plexus

Cont…

- Bacterial survival in the sub-arachnoid space
  - Low complement and immunoglobulin concentration
- Inflammation in the sub-arachnoid space
  - Interleukins, IL-1β, IL-6,IL-8, Interferon -ý
  - TNF-α,
  - PGE2
  - Nitric oxide

Cont…

- Alteration of the Blood-Brain barrier
  - Separation of intercellular tight junctions
  - Increased pinocytic vesicles
- Increased intracranial pressure
  - Cerebral edema
- Alteration in cerebral blood flow
  - Microvascular vasulitis, thrombosis
  - Cerebral ishcemia/infarction
- Neuronal injury
  - Apoptotic neuronal injury

Epidemiology

- Approximately 1.2 million cases of bacterial meningitis occur annually worldwide
- Meningitis is among the ten most common infectious causes of death and is responsible for approximately 135,000 deaths throughout the world each year
- Neurologic sequelae are common among survivors

Cont…

- Epidemic meningococcal disease in African Meningitis Belt
  - Epidemics occur every 5- 10 years
  - Very large epidemics with high attack rates
  - Epidemics occur in dry season and decrease in the rainy season
  - Peak age group 5-14 years
  - Epidemics caused by serogroup A, less frequently serogroup C
  - Epidemic thresholds (WHO, 1998)
    - 15 cases/100,000 population/week (Population>30,000)
    - 5 cases in a week (Population<30,000)

Etiology

- The major causes of community-acquired bacterial meningitis in adults are Streptococcus pneumoniae, Neisseria meningitidis
- Listeria monocytogenes primarily in patients over age 50 to 60 years or those who have deficiencies in cell-mediated immunity
- S. pneumoniae is the most common cause of meningitis in adults >20 years of age, accounting for nearly half the reported cases
- N. meningitidis accounts for 25% of all cases of bacterial meningitis

Cont…

- Enteric gram-negative bacilli are an increasingly common cause of meningitis in individuals with chronic and debilitating diseases such as diabetes, cirrhosis, or alcoholism and in those with chronic urinary tract infections
- Gram-negative meningitis can also complicate neurosurgical procedures, particularly craniotomy
- The major causes of healthcare-associated bacterial meningitis are staphylococci and aerobic gram-negative bacilli

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Clinical Features

- Patients with bacterial meningitis are usually quite ill and often present soon after symptom onset
- The classic triad of acute bacterial meningitis consists of fever, nuchal rigidity, and a change in mental status
- Nuchal rigidity ("stiff neck") is the pathognomonic sign of meningeal irritation and is present when the neck resists passive flexion
- A decreased level of consciousness occurs in >75% of patients and can vary from lethargy to coma
- Nausea, vomiting, and photophobia are also common complaints

Cont…

- The sensitivity and specificity of Kernig's and Brudzinski's signs are uncertain
- Both may be absent or reduced in very young or elderly patients, immunocompromised individuals, or patients with a severely depressed mental status
- The high prevalence of cervical spine disease in older individuals may result in false-positive tests for nuchal rigidity

Cont…

- The classic Brudzinski's sign refers to spontaneous flexion of the hips during attempted passive flexion of the neck.
- The Kernig's sign refers to the inability or reluctance to allow full extension of the knee when the hip is flexed 90º.
- Kernig's test is usually performed in the supine position, but it can be tested in the seated patient.

Cont…

- Seizures occur as part of the initial presentation of bacterial meningitis or during the course of the illness in 20–40% of patients
- Raised ICP is an expected complication of bacterial meningitis and the major cause of obtundation and coma in this disease

Cont…

Certain bacteria, particularly N. meningitidis, can cause characteristic skin manifestations, such as petechiae and palpable purpura



Laboratory findings

- Approximately 50 to 90 percent of patients with bacterial meningitis have positive blood cultures
- Every patient with suspected meningitis should have CSF obtained unless a lumbar puncture (LP) is contraindicated
- computed tomographic (CT) scan is performed to exclude a mass lesion or increased intracranial pressure

Cont…

- The usual CSF findings in patients with bacterial meningitis are a white blood cell count of 1000 to 5000/microL with a percentage of neutrophils usually greater than 80 percent
- Protein of 100 to 500 mg/dL
- Glucose <40 mg/dL (CSF: serum glucose ratio of <0.4)
- A Gram stain should be obtained whenever there is suspicion of bacterial meningitis

Diagnosis, CSF Examination

CSF parameter	**Typical finding**
Opening pressure	200-500mmH2O
White blood cell count	1000-5000/mm3; %Neutrophils >80%
Protein	100-500mg/dl
Glucose	<40mg/dl; CSF:serum glucose<0.4
Gram Stain	Positive in 60-90%
Culture	Positive in 70-85%

Cont…

Immunocompromised state (eg, HIV infection, immunosuppressive therapy, solid organ or hematopoietic stem cell transplantation)

History of CNS disease (mass lesion, stroke, or focal infection)

New onset seizure (within one week of presentation)

Papilledema

Abnormal level of consciousness

Focal neurologic deficit

Cont…

$$\text{True WBC in CSF} = \text{Actual WBC in CSF} - \frac{\text{WBC in blood} \times \text{RBC in CSF}}{\text{RBC in blood}}$$

Cont…

- Gram-positive diplococci suggest pneumococcal infection
- Gram-negative diplococci suggest meningococcal infection
- Small pleomorphic gram-negative coccobacilli suggest Haemophilus influenzae infection
- Gram-positive rods and coccobacilli suggest listerial infection

Initial therapy/Empiric antibiotics

- The mortality rate of untreated disease approaches 100 percent and, even with optimal therapy, there is a high failure rate
- There are three general requirements of antibiotic therapy for bacterial meningitis:
  1. use of bactericidal drugs effective against the infecting organism
  2. use of drugs that enter the CSF
  3. use of drugs with optimal pharmacodynamic

Cont…

- Antibiotic therapy must be initiated immediately after the performance of the lumbar puncture (LP) or, if a computed tomography scan is to be performed before LP, immediately after blood cultures are obtained
- Adjunctive dexamethasone should be given shortly before or at the same time as the first dose of antibiotics, when indicated

Cont…

- No known immunodeficiency
  - Streptococcus pneumoniae, Neisseria meningitidis, and, less often, Haemophilus influenzae and

group B streptococcus are the most likely causes of community-acquired bacterial meningitis

  - Individuals aged 50 years and older are also at substantial risk of Listeria monocytogenes meningitis

Cont…

- Ceftriaxone— 2 g IV every 12 hours *OR*
- Cefotaxime— 2 g IV every four to six hours

  • PLUS

- Vancomycin— 30 to 60 mg/kg IV per day in two or three divided doses

  • PLUS

- In adults ≥50 years of age, ampicillin— 2 g IV every four hours

Cont…

- Impaired cell-mediated immunity
  - Among patients with impaired cell-mediated immunity, coverage must be directed against Listeria monocytogenes and gram-negative bacilli (including Pseudomonas aeruginosa) as well as Streptococcus pneumoniae

Cont…

- An appropriate regimen in patients with normal renal function is:
  - Vancomycin— 30 to 60 mg/kg IV per day in two or three divided doses
  - PLUS
    - Ampicillin— 2 g IV every four hours
  - PLUS EITHER
    - Cefepime— 2 g IV every eight hours
  - OR
    - Meropenem— 2 g IV every eight hours — 2 g IV every eight hours

Cont…

- Nosocomial meningitis
  - Empiric therapy for nosocomial meningitis must cover both gram-positive and gram-negative (such as Klebsiella pneumoniae and Pseudomonas aeruginosa) nosocomial pathogens
  - Vancomycin— 30 to 60 mg/kg IV per day in two or three divided doses
  - PLUS
    - Ceftazidime— 2 g IV every eight hours
  - OR
    - Cefepime— 2 g IV every eight hours
  - OR
  - Meropenem — 2 g IV every eight hours

Empiric therapy for bacterial meningitis

Predisposing factor	**Antimicrobial therapy**
Age <1 month	Ampicillin + a third generation cephalosporin
Age 1-23 months	Vancomycin + a third generation cephalosporin
Age 2-50 years	Vancomycin + a third generation cephalosporin
Age > 50 years	Vancomycin + Ampicillin + a third generation cephalosporin
head trauma; postneurosurgey	Vancomycin + ceftazidime or cefepime
Basilar skull fracture	Vancomycin + a third generation cephalosporin
Immunocompromised state	Vancomycin + Ampicillin + ceftazidime or cefepime

Prevention

- Vaccines
  - Meningicoccal vaccine
  - Should be administered to any asplenic patient
- Exposure to meningococcus
  - Rifampin 600 mg PO BID x 4 doses
  - Only for intimate contacts: spouse, boyfriend/girlfriend, household contacts

Acute Viral Meningitis

Viral Meningitis

- 75% caused by enteroviruses
  - Enterovirus
  - Coxsackie virus
  - Echo virus
- Other viruses
  - HSV2 (HSV1 causes encephalitis)
  - HIV
  - Lymphocytic choriomeningitis virus
  - Mumps
  - Varicella Zoster

Viral Meningitis

- Cannot distinguish initially from bacterial meningitis
- Severe HA, photophobia, nucal rigidity, fever
- It may be preceded by a few weeks by viral gastroenteritis
- Almost never involves brain (meningoencephalitis)
  - Pt never obtunded, no Hx of seizure
- Disease is self-limited, resolves after 7 to 10 days without treatment
- No serious sequelae

CSF

- Low numbers of WBCs : 10 to 500
  - PMNs predominate early, Monos or Lymphocytes later
- CSF to serum glucose ratio usually = 50%
- Protein may be high
- Gram stain, culture and bacterial antigens negative
- Enteroviral PCR positive about 70% of time

Approach to Viral Meningitis

- Treat like bacterial meningitis until the 72 hr culture comes back negative, or…
- Enteroviral PCR comes back positive
- Consider acyclovir if CSF HSV PCR positive
  - HSV meningitis is self-limited

Viral Encephalitis

Cont…

- **Definition**
  - brain parenchyma is involved
- Patients with aseptic meningitis
  - most commonly present with fever and headache with meningismus on examination
  - may be lethargic but have a normal sensorium
- Patients with encephalitis present with mental status changes
- Patients with features of both may be considered to have a meningoencephalitis

Cont…

- Obtain neuroimaging to assess the possibility of a localized process, such as an abscess, and to look for focality, which may suggest a specific etiology of encephalitis (eg, temporal lobe involvement and HSV-1)
- A lymphocytic predominance is suggestive of a viral etiology; red blood cells in the absence of a traumatic tap is suggestive of HSV or other necrotizing viral encephalitides
- The most important diagnostic consideration in patients with encephalitis is HSV since delay in treatment can impact morbidity and mortality

Cont…

- The most important viruses causing sporadic cases of encephalitis in immunocompetent adults are herpesviruses (HSV, VZV, EBV)
- The characteristic CSF profile is indistinguishable from that of viral meningitis and typically consists of a lymphocytic pleocytosis, a mildly elevated protein concentration, and a normal glucose concentration

Cont…

- A CSF pleocytosis (>5 cells/L) occurs in >95% of patients with documented viral encephalitis
- CSF cell counts exceed 500/L in only about 10% of patients with encephalitis
- A decreased CSF glucose concentration is distinctly unusual in viral encephalitis and should suggest other Dx

Cont…

- Dx

✓CSF PCR

✓Serology & Ag detection

- Treatment
  - Specific antiviral
    - Acyclovir
    - Foscarne
    - Ganciclovir

Quiz

1.List at least three cause and management of constipation?

2. Write clinical classification of dehydration based on the cause of diarrheal ?